AIM :

To write a Python program to implement the Tic Tac Toe game between two players.

ALGORITHM :

1. Start the program.
2. Initialize an empty 3×3 board.
3. Display the board.
4. Get the position from the current player.
5. Check whether the position is valid.
6. Place X or O in the selected position.
7. Check whether the player has won.
8. If a player wins, display the winner.
9. Otherwise, switch to the next player.
10. If all positions are filled, declare a draw.

11. Stop the program.

PROGRAM :

prgm 12.py - C:/Users/nivet/AppData/Local/Programs/Python/Python314/prgm 12.py (3.14.6)

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```
# Tic Tac Toe Game

board = [' ' for i in range(9)]

def display():
    print()
    print(board[0], '|', board[1], '|', board[2])
    print('--+---+--')
    print(board[3], '|', board[4], '|', board[5])
    print('--+---+--')
    print(board[6], '|', board[7], '|', board[8])
    print()

def check_winner(player):
    win = [
        [0, 1, 2], [3, 4, 5], [6, 7, 8],
        [0, 3, 6], [1, 4, 7], [2, 5, 8],
        [0, 4, 8], [2, 4, 6]
    ]

    for line in win:
        if all(board[i] == player for i in line):
            return True
    return False

def play():
    player = 'X'

    for turn in range(9):
        display()
        pos = int(input("Enter position (1-9): ")) - 1

        if pos < 0 or pos > 8 or board[pos] != ' ':
            print("Invalid move! Try again.")
            continue

        board[pos] = player

        if check_winner(player):
            display()
            print(player, "wins!")
            return

        player = 'O' if player == 'X' else 'X'
    display()
    print("It's a Draw!")
play()
```

OUTPUT :

```
========== RESTART: C:/Users/nivet/AppData/Local/Programs/Python/Python314/prgm 12.py ==========


  |   |
--+---+--
  |   |
--+---+--
  |   |


Enter position (1-9):
```

RESULT :

Thus, the Tic Tac Toe game was successfully implemented using Python.